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Sniffing Cleartext Protocols Lab:

To start off for this lab, we will need to open Wireshark (which is pre-installed with Kali) and prepare a text file to be sent over FTP.

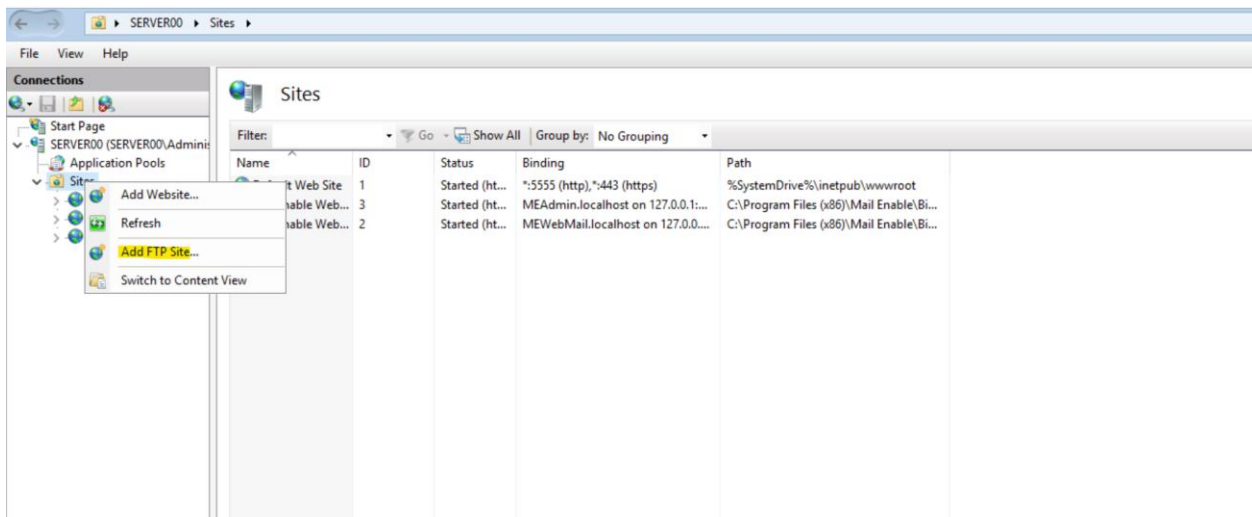
To do this, I simply opened the terminal and entered “cat > hello.txt”, and then entered “hello world”, and saved the file.

```
(kali@kali):~$ cat > hello.txt
hello world
^C
(kali@kali):~$ ls
Desktop  Documents  Downloads  hello.txt  http_mcdonalds.com_robots.txt  list-of-ips.txt  mcdonaldsreport.html  Music  myfile.txt  nsescripts.txt  Pictures  Public  Templates  Videos
```

Next, if ftp is not installed, you will have to do so. For me, FTP was already installed on my Kali machine.

```
(root@kali)-[/home/kali]
# apt install ftp
ftp is already the newest version (20230507-2).
ftp set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1314
```

Before continuing, we will need to set up the FTP server on Windows Server, which is not yet configured. To do this, open the IIS manager, and on the “sites” option, click “ADD FTP SITE”.



Next, we will need to give the site a name and set up a directory for its contents to be stored in it. As pictured below, I named the site “FTP Site” and set the path in the C drive.

**Site Information**

FTP site name:

FTP Site

Content Directory

Physical path:

C:\



Previous

Next

Finish

Cancel

On the next page, the only thing that will need to be changed is the SSL option. Simply change this to “No SSL”.

**Binding and SSL Settings**

Binding

IP Address: Port:

☐ Enable Virtual Host Names:

Virtual Host (example: ftp.contoso.com):

☒ Start FTP site automatically

SSL

☒ No SSL

☐ Allow SSL

☐ Require SSL

SSL Certificate:

Finally, on the last page check off “Basic”, and then allow access to “Specified roles or user groups”, type “administrator” in the box, check off the read and write permissions, and then finally hit finish.

Add FTP Site ? X

 **Authentication and Authorization Information**

Authentication

☐ Anonymous

☒ Basic

Authorization

Allow access to:

Specified roles or user groups

administrator

Permissions

☒ Read

☒ Write

Previous Next Finish Cancel

If we check the sites tab now, we can see our FTP server is up and running.

File View Help

Connections

- Start Page
- SERVER00 (SERVER00\Admini...
- Application Pools
- Sites
 - Default Web Site
 - MailEnable WebAdmin
 - MailEnable WebMail
 - FTP Site

Sites

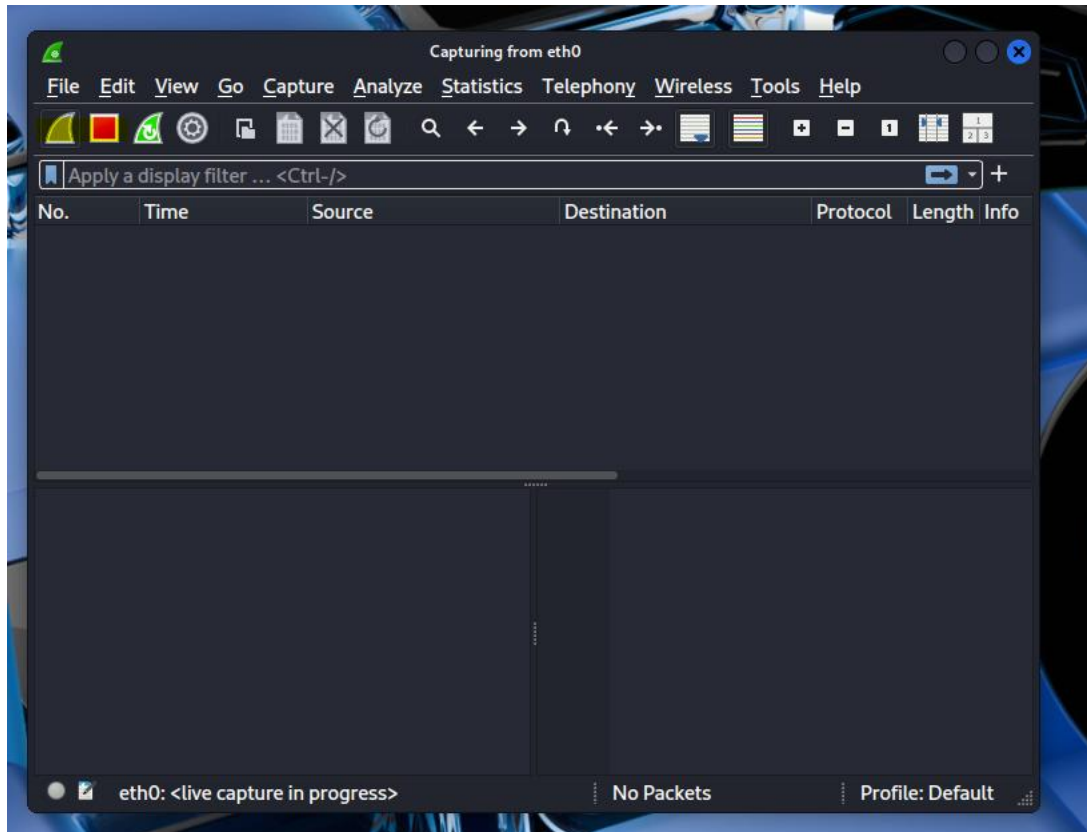
Filter: Show All Group by: No Grouping

Name	ID	Status	Binding	Path
Default Web Site	1	Started (ht...	*:5555 (http),*:443 (https)	%SystemDrive%\inetpub\wwwroot
FTP Site	4	Started (ftp)	*:21: (ftp)	C:\
MailEnable Web...	3	Started (ht...	MEAdmin.localhost on 127.0.0.1:...	C:\Program Files (x86)\Mail Enable\Bi...
MailEnable Web...	2	Started (ht...	MEWebMail.localhost on 127.0.0.0...	C:\Program Files (x86)\Mail Enable\Bi...

Now if we try to use FTP from the Kali machine to the Windows server, we are able to login and connect with no issues.

```
REDACTED
REDACTED
REDACTED administrator
Password:
230 User logged in.
Remote system type is Windows_NT.
ftp>
```

Next, we will prepare the “hello.txt” file for transfer via FTP, but first we will need to start Wireshark, and start capturing packets.



Once we have started Wireshark, on the FTP session enter command “put hello.txt” and then confirm the transfer was complete with the “Transfer complete” message as highlighted below.

```
ftp> put hello.txt
local: hello.txt remote: hello.txt
229 Entering Extended Passive Mode (|||49677|)
125 Data connection already open; Transfer starting.
100% |*****| 14 651.04 KiB/s --:-- ETA
226 Transfer complete.
14 bytes sent in 00:00 (17.75 KiB/s)
ftp>
```

After transferring the file, on Wireshark stop capturing packets and apply a search filter for FTP. Once applied, you should get an output like the one below:

No.	Time	Source	Destination	Protocol	Length	Info
986	173.785165184	192.168.116.129	192.168.116.128	FTP	93	Response: 220 Microsoft FTP Service
988	176.483481728	192.168.116.128	192.168.116.129	FTP	86	Request: USER administrator
989	176.484028964	192.168.116.129	192.168.116.128	FTP	89	Response: 331 Password required
991	180.162866560	192.168.116.128	192.168.116.129	FTP	81	Request: PASS Pa22w0rd
992	180.163877346	192.168.116.129	192.168.116.128	FTP	87	Response: 230 User logged in.
994	180.164833007	192.168.116.128	192.168.116.129	FTP	72	Request: SYST
995	180.164364837	192.168.116.129	192.168.116.128	FTP	82	Response: 215 Windows_NT
996	180.164415832	192.168.116.128	192.168.116.129	FTP	72	Request: FEAT
997	180.164798315	192.168.116.129	192.168.116.128	FTP	100	Response: 211-Extended features supported:
998	180.164798408	192.168.116.129	192.168.116.128	FTP	84	Response: LANG EN*
999	180.164798430	192.168.116.129	192.168.116.128	FTP	119	Response: AUTH TLS;TLS-C;SSL;TLS-P;
1000	180.164798451	192.168.116.129	192.168.116.128	FTP	73	Response: HOST
1001	180.164828200	192.168.116.129	192.168.116.128	FTP	103	Response: SIZE
1167	192.747821019	192.168.116.129	192.168.116.128	FTP	72	Request: EPSV
1168	192.748555476	192.168.116.129	192.168.116.128	FTP	114	Response: 229 Entering Extended Passive Mode (49677)
1172	192.748996390	192.168.116.128	192.168.116.129	FTP	82	Request: STOR hello.txt
1173	192.749864557	192.168.116.129	192.168.116.128	FTP	120	Response: 125 Data connection already open; Transfer starting.
1178	192.750626238	192.168.116.129	192.168.116.128	FTP	90	Response: 226 Transfer complete.

Now, for any one of the packets, right click and hit “Follow”, and then “TCP stream”. This will reveal the entire session of events, including the username and password.

```

220 Microsoft FTP Service
USER administrator
331 Password required
PASS Pa22w0rd
230 User logged in.
SYST
215 Windows_NT
FEAT
211-Extended features supported:
LANG EN*
UTF8
AUTH TLS;TLS-C;SSL;TLS-P;
PBSZ
PROT C;P;
CCC
HOST
SIZE
MDTM
REST STREAM
211 END
EPSV
229 Entering Extended Passive Mode (|||49677|)
STOR hello.txt
125 Data connection already open; Transfer starting.
226 Transfer complete.

```

Next, filter out packets for “tcp-data”. As pictured below, there is only one packet with data available to view:

No.	ftp-data	Source	Destination	Protocol	Length	Info
1174	192.749981077	192.168.116.128	192.168.116.129	FTP-DA	80	FTP Data: 14 bytes (EPSV) (STOR hello.txt)

Next, click on the packet to view its contents. If done correctly, we can see the text that I had entered previously, “hello world” in the bottom left corner.

```

* Frame 1174: 88 bytes on wire (648 bits), 88 bytes captured (648 bits) on interface eth0, id 0
* Ethernet II, Src: VMware 59:cf:41 (00:0c:29:59:cf:41), Dst: VMware 3c:32:6c (00:0c:29:3c:32:6c)
* Internet Protocol Version 4, Src: 192.168.116.128, Dst: 192.168.116.129
* Transmission Control Protocol, Src Port: 37486, Dst Port: 49677, Seq: 1, Ack: 1, Len: 14
  FTP Data (14 bytes data)
    [Setup frame: 1168]
    [Setup method: EPASV]
    [Command: STOR hello.txt]
    [Current working directory: ]
    [Command frame: 1172]
    [Current working directory: ]
    Line-based text data (1 lines)

```

```

0000  00 0c 29 3c 32 6c 00 0c 29 59 cf 41 00 00 45 0a  ]<21..}Y A. E.
0010  00 42 1a 3b 40 00 40 06 b6 20 c0 a9 74 00 c0 a8  8 ;@ @ . t .
0020  74 01 92 6e c2 00 17 7e 00 0c 29 f9 72 f7 00 15  t h . . # f .
0030  7e e3 6a 87 00 00 01 01 00 0a f2 0c d7 00 00 16  - j . . . . l . .
0040  23 f2 68 65 6c 6c 6f 20 77 6f 72 0c 64 21 0d 0a  # hello world!

```